InsightRAG — Project Handover Document

Course: CPSC 319
Team: Sherry Rui Xia, Jaskirat Gill, and team
Handover Date: April 20, 2025
Project: InsightRAG — Knowledge Base Control Plane for RAG Systems

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Quick Status Overview

Status	Features
Fully working	Authentication & RBAC, Knowledge Base management, Document sync from S3, Document processing pipeline (parse → chunk → embed → index), Hybrid search (vector + keyword), MCP server with 3 tools, Retrieval analytics & heatmaps, Health dashboard, User management, CI/CD pipeline
Partially working	Document strategy override (saves setting but does not re-trigger processing), Real-time S3 sync via SQS (works if SQS is configured; falls back to polling otherwise)
Not implemented	Chat interface (UI component exists but not accessible), Additional sync plugins (only S3 is supported), Rate limiting, Observability / metrics (Prometheus, Jaeger)

Demo Access

Use these credentials to log in to any running instance — no setup required.

Field	Value
Email	`admin@example.com`
Password	`Admin123!`

- **Local (after setup):** http://localhost:5173

The database is automatically seeded with this admin account on first start. You do not need to create a user manually.

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1. Project Overview

InsightRAG is a knowledge base control plane for Retrieval-Augmented Generation (RAG) systems. It allows teams to:

- **Sync** documents from external cloud storage (AWS S3) into a managed index via a plugin system
- **Process** documents through a pipeline: parse → chunk → generate embeddings → index into a Qdrant vector database
- **Search** indexed content using hybrid retrieval (vector + keyword search with RRF fusion)
- **Expose** search capabilities via a Model Context Protocol (MCP) server, compatible with Claude and other AI assistants
- **Monitor** knowledge base and document health, retrieval analytics, and chunk-level statistics through a web UI
- **Manage** users, roles, and API keys with a role-based access control (RBAC) system

Built-in documentation: Visit `http://localhost:5173/docs` on a running local instance for the interactive docs site covering Quick Start, Features, Plugin Setup, MCP Setup, and team information.

GitHub repository: https://github.com/jaskirat-gill/InsightRAG

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2. Technology Stack

Layer	Technology
Frontend	React 19, Vite 7, TypeScript, Tailwind CSS, Radix UI
Sync Service (API)	Python 3.11, FastAPI 0.109, SQLModel, Celery

Layer	Technology
Document Processing	Python 3.11, Celery, sentence-transformers, unstructured
Query Engine	Python 3.11, FastAPI 0.109
MCP Server	Python 3.11, FastMCP 3.0
Vector Database	Qdrant
Relational Database	PostgreSQL 15
Cache / Task Queue	Redis 7
Containerization	Docker, Docker Compose
Reverse Proxy	nginx (production)
CI/CD	GitHub Actions → DigitalOcean droplet
Node.js	v22 (frontend build)

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3. Prerequisites

Before setting up the project locally, ensure the following are installed:

Tool	Minimum Version	Download
Docker Desktop	24+	https://www.docker.com/products/docker-desktop
Docker Compose	v2 (bundled with Docker Desktop)	(included)
Node.js	22+	https://nodejs.org
npm	10+ (bundled with Node.js)	(included)
Git	Any recent version	https://git-scm.com

Optional (for MCP integration):

- An AWS account with an S3 bucket (see Step 3)
- Claude Desktop or another MCP-compatible client

Verify your installations by running:

```
docker --version
docker compose version
node --version
npm --version
git --version
```

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4. Step-by-Step Setup Guide

Step 1: Clone the Repository

Open your terminal and run:

```
git clone https://github.com/jaskirat-gill/InsightRAG.git
cd InsightRAG
```

Step 2: Configure Environment Variables

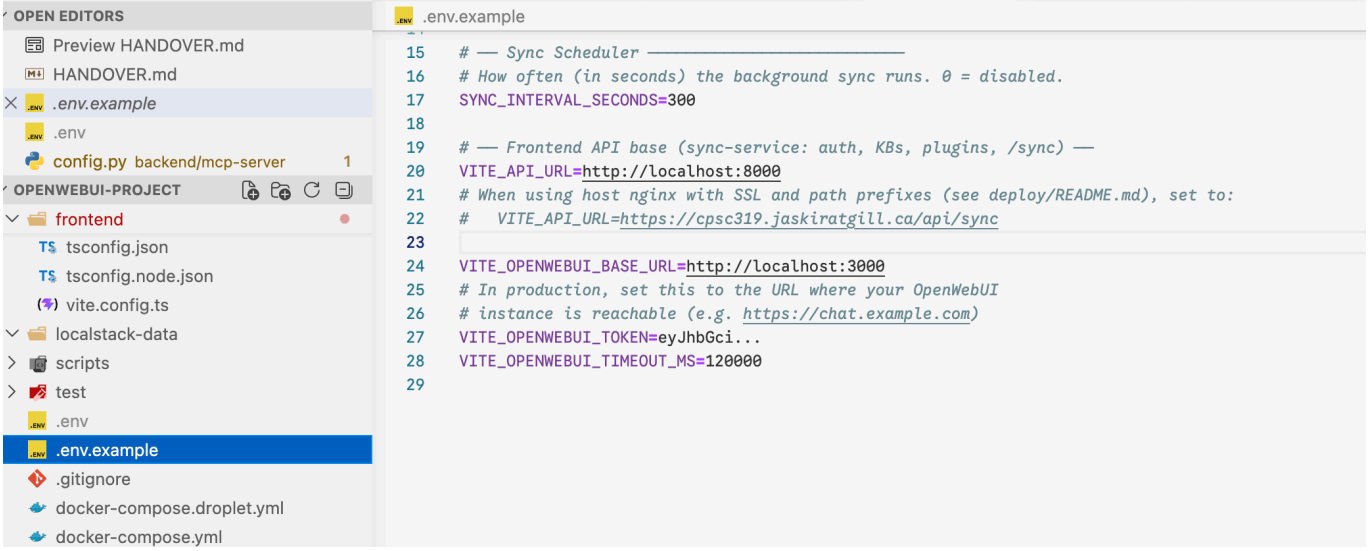
The project uses a `.env` file to store configuration. A template is provided.

```
cp .env.example .env
```

Open `.env` in a text editor and fill in the required values:

```
# ── AWS / S3 (required for document sync) ──────────────────
AWS_ACCESS_KEY_ID=AKIA...          # Your AWS Access Key ID
AWS_SECRET_ACCESS_KEY=wJalr...     # Your AWS Secret Access Key
AWS_REGION=us-east-1               # Your S3 bucket region
S3_BUCKET_NAME=your-bucket-name    # Your S3 bucket name

# ── SQS (required — must be a valid queue URL, see Step 3e) ──
SQS_QUEUE_URL=https://sqs.us-east-1.amazonaws.com/123456789012/insightrag-
s3-events

# ── Sync Scheduler ─────────────────────────────────────────
# How often background sync runs (seconds). Set to 0 to disable.
SYNC_INTERVAL_SECONDS=300

# ── Frontend API Base ──────────────────────────────────────
# For local development, leave as-is:
VITE_API_URL=http://localhost:8000

# ── OpenWebUI Integration (optional) ───────────────────────
VITE_OPENWEBUI_BASE_URL=http://localhost:3000
VITE_OPENWEBUI_TOKEN=eyJhbGci...
VITE_OPENWEBUI_TIMEOUT_MS=120000
```

> **Important:** Never commit your `.env` file to version control. It is already listed in `.gitignore`.



Step 3: Set Up AWS S3 Credentials

If you want to use the document sync feature, you need an AWS S3 bucket and IAM credentials.

3a. Create an S3 Bucket

1. Go to the Amazon S3 Console
2. Click **Create bucket**
3. Enter a bucket name (e.g., `insightrag-docs`) and select a region
4. Leave **Block Public Access** enabled
5. Click **Create bucket**

aws Search [Option+S] United States (Oh 2629172636@qq.com (2731-3055-8057) 2629172636@qq.com

Amazon S3 > Buckets > Create bucket

Create bucket Info

Buckets are containers for data stored in S3.

General configuration

AWS Region

US East (Ohio) us-east-2

Bucket type Info

- (●) General purpose
  Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
- ( ) Directory
  Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace

Choose the namespace where you want to create your bucket. Learn more

- (●) Global namespace
  By default, S3 creates general purpose buckets in the global namespace.
- ( ) Account Regional namespace (recommended)
  General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name Info

my-knowledge-base-bucket-1

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). Learn more

Copy settings from existing bucket - *optional*

Only the bucket settings in the following configuration are copied.

Choose bucket

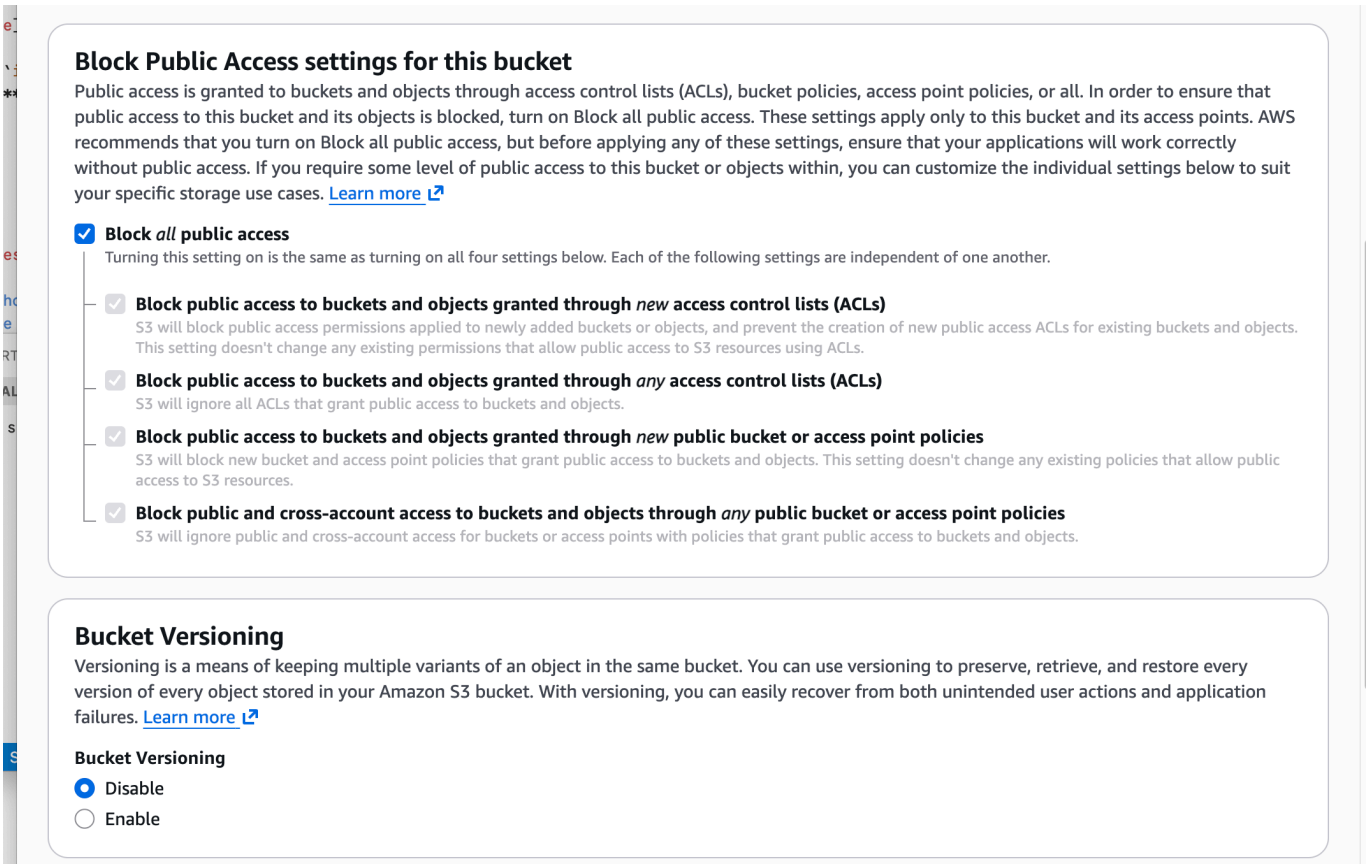
Format: s3://bucket/prefix

Object Ownership Info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

- (●) ACLs disabled (recommended)
- ( ) ACLs enabled

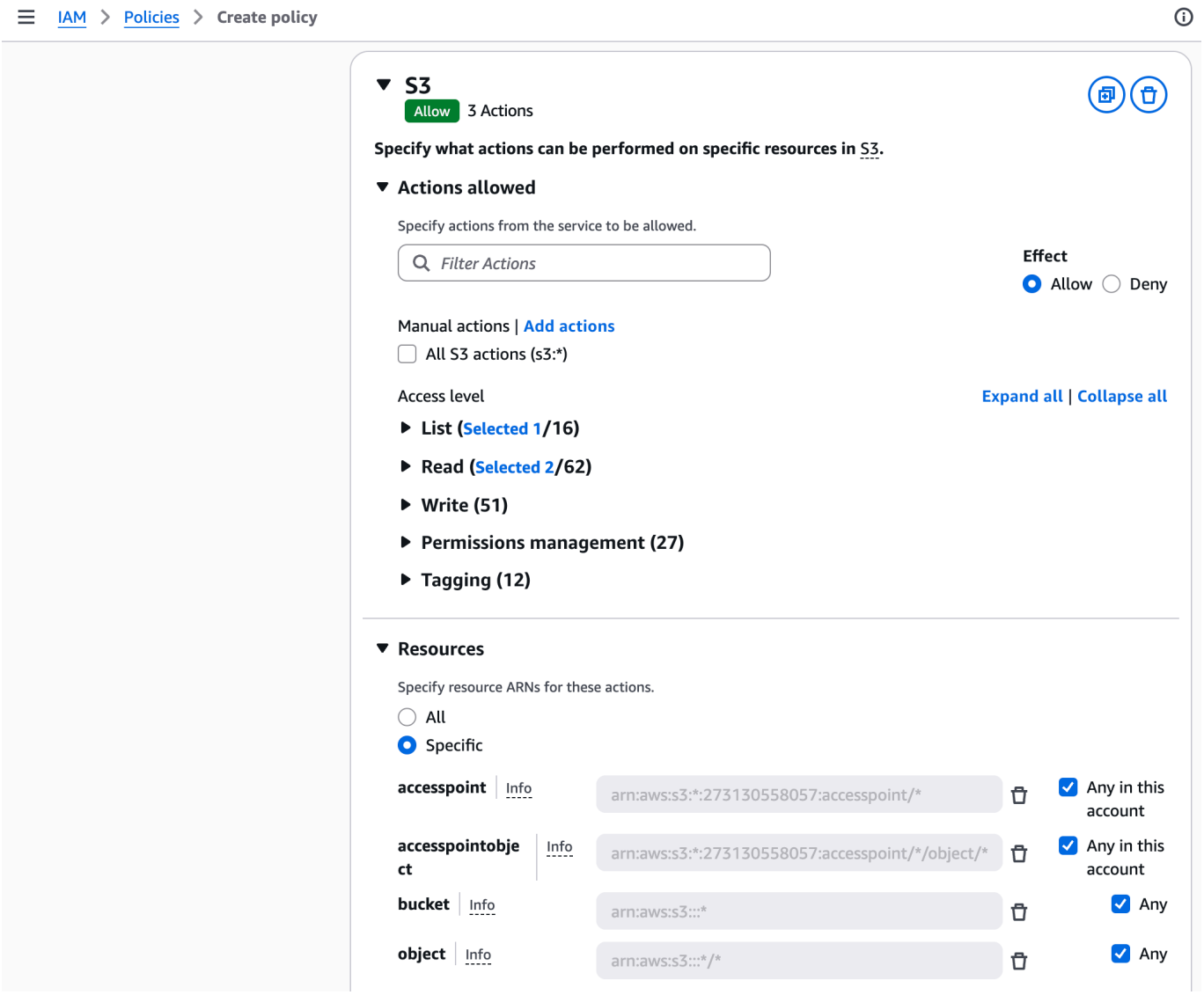


3b. Create an IAM Policy

1. Go to IAM Console → Policies
2. Click **Create policy** → choose **JSON**
   - or use IAM console ui, choose S3 -> List Bucket, GetObject, GetBucketLocations
3. Paste the following (replace `YOUR_BUCKET_NAME`):

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "s3:ListBucket",
                "s3:GetObject",
                "s3:GetBucketLocation"
            ],
            "Resource": [
                "arn:aws:s3:::YOUR_BUCKET_NAME",
                "arn:aws:s3:::YOUR_BUCKET_NAME/*"
            ]
        }
    ]
}
```

4. Name the policy (e.g., `InsightRAG-S3-Read`) and click **Create policy**



3c. Create an IAM User

1. Go to IAM Console → Users
2. Click **Create user** → enter a name (e.g., `insightrag-sync`)
3. Select **Attach policies directly** → find and select `InsightRAG-S3-Read` & `AmazonSQSFullAccess`
4. Click **Create user**

IAM > Users

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management

Users (1) Info

Delete | Create user

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

User name	Path	Groups	Last activity
openwebui-sync	/	0	10 days ago

IAM > Users > Create user

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. Learn more

Permissions options

- Add user to group — Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.
- Copy permissions — Copy all group memberships, attached managed policies, and inline policies from an existing user.
- Attach policies directly — Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (2/1485)

Create policy

Choose one or more policies to attach to your new user.

Filter by Type

insightRa | All types | 1 match

Policy name	Type	Attached entities
InsightRAG-S3-Read	Customer managed	0

Set permissions boundary - *optional*

Cancel | Previous | Next

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name	Console password type	Require password reset
insightrag-sync	None	No

Permissions summary

1

Name	Type	Used as
AmazonSQSFullAccess	AWS managed	Permissions policy
InsightRAG-S3-Read	Customer managed	Permissions policy

Tags - *optional*

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

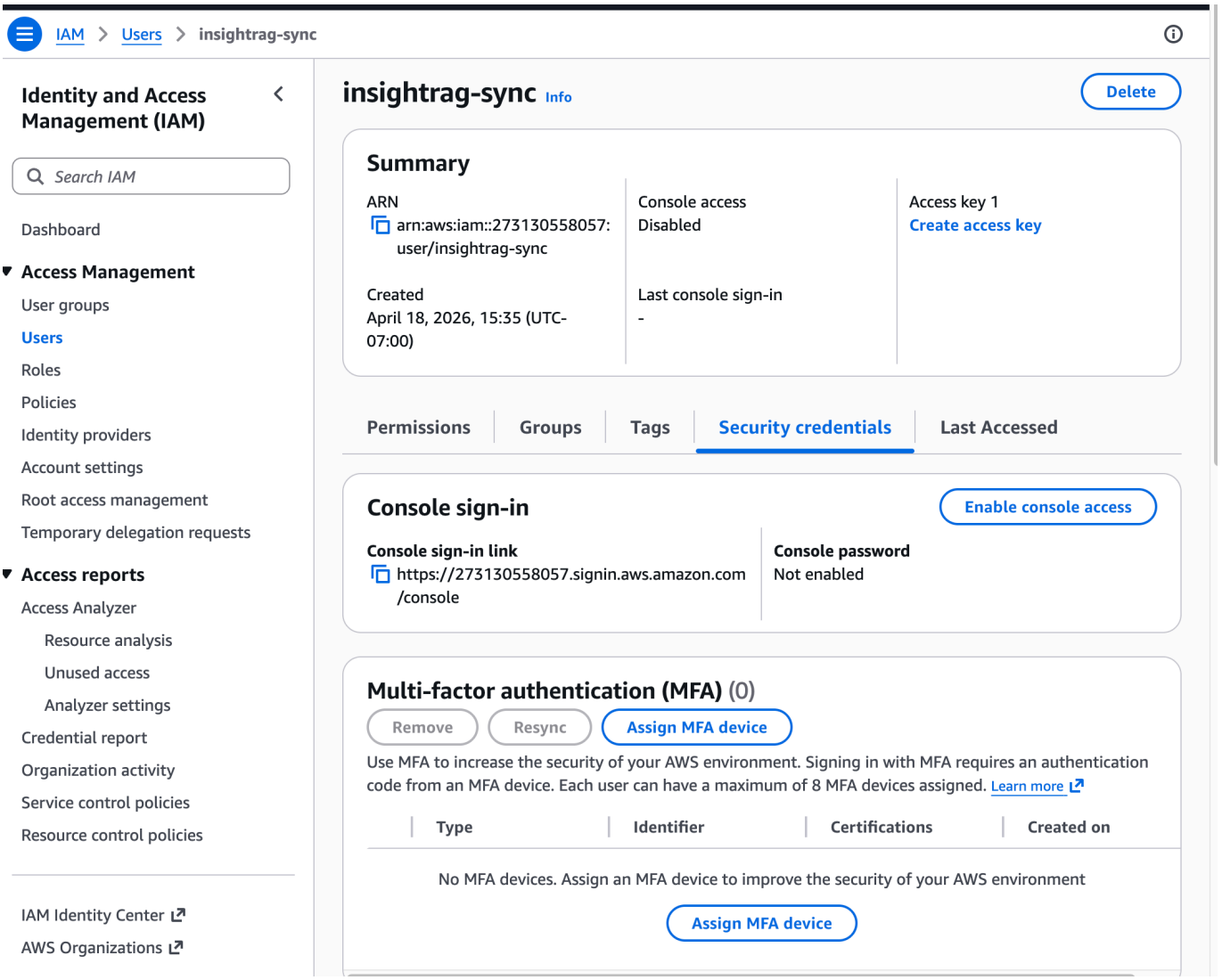
No tags associated with the resource.

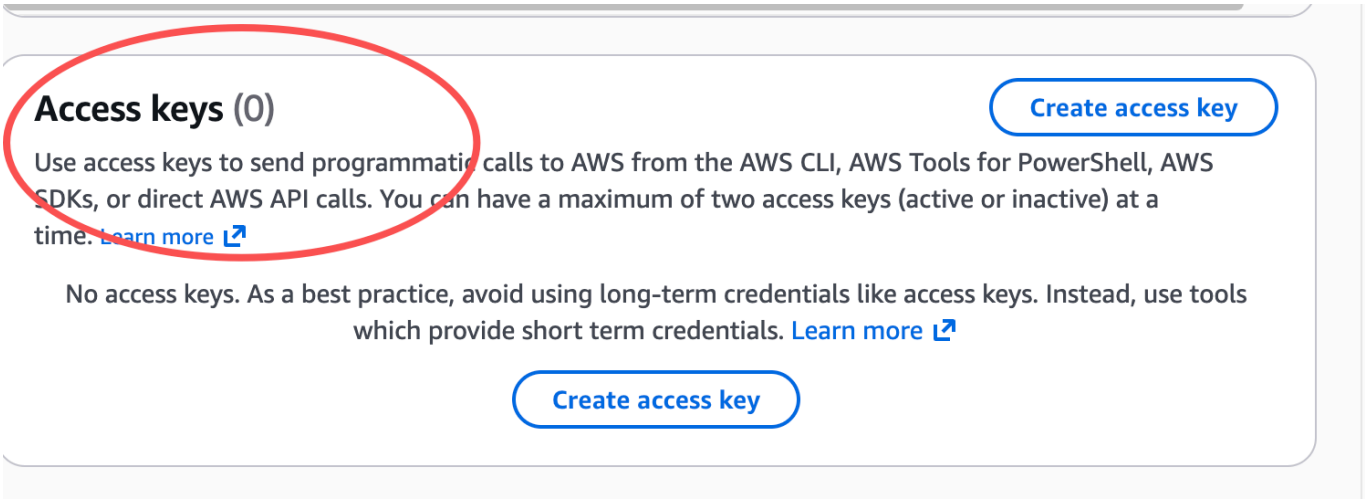
Add new tag

You can add up to 50 more tags.

3d. Generate Access Keys

1. Click on the new user → go to **Security credentials** tab
2. Scroll to **Access keys** → click **Create access key**
3. Select **Application running outside AWS**
4. Copy the **Access Key ID** and **Secret Access Key** into your `.env` file





3e. Create an SQS Queue and Wire It to S3 (Required)

`SQS_QUEUE_URL` **must** be a valid SQS queue URL. Leaving it empty or using a placeholder causes the sync to crash with an `InvalidAddress` error.

1. Go to the Amazon SQS Console
2. Click **Create queue** → choose **Standard** (not FIFO)
3. Name it (e.g., `insightrag-s3-events`)
4. Set **Receive message wait time** to **5 seconds** (enables long polling)

5. Click **Create queue**
6. Copy the **Queue URL** — it looks like `https://sqs.us-east-1.amazonaws.com/123456789012/insightrag-s3-events`

Allow S3 to send messages to this queue — on the SQS queue page, go to **Access policy** and add this statement (replace the three placeholders):

```
{
    "Effect": "Allow",
    "Principal": { "Service": "s3.amazonaws.com" },
    "Action": "SQS:SendMessage",
    "Resource": "arn:aws:sqs:*:YOUR_AWS_ACCOUNT_ID:YOUR_QUEUE_NAME",
    "Condition": {
        "ArnLike": { "aws:SourceArn": "arn:aws:s3:::YOUR_BUCKET_NAME" }
    }
}
```

Configure S3 event notifications — in your S3 bucket:

1. Go to **Properties → Event notifications → Create event notification**
2. Under **Event types** select `s3:ObjectCreated:*` and `s3:ObjectRemoved:*`
3. Under **Destination** choose **SQS queue** and select the queue you just created
4. Click **Save changes**

Add SQS permissions to the IAM policy you created in step 3b:

```
{
    "Effect": "Allow",
    "Action": [
        "sqs:ReceiveMessage",
        "sqs:DeleteMessage",
        "sqs:GetQueueAttributes"
    ],
    "Resource": "arn:aws:sqs:*:YOUR_AWS_ACCOUNT_ID:YOUR_QUEUE_NAME"
}
```

Add the Queue URL to `.env`:

```
SQS_QUEUE_URL=https://sqs.us-east-1.amazonaws.com/123456789012/insightrag-s3-events
```

Step 4: Start Backend Services

From the project root directory, run:

```
docker compose up --build
```

This command builds and starts the following 8 services:

Service	Description	Host Port
`sync-service`	Core API: auth, KB management, plugins, sync	`8000`
`query-engine`	Search/query API used by the UI	`8001`
`mcp-server`	MCP server for AI assistant integration	`8002`
`document-processing-engine`	Celery worker: parse, chunk, embed, index	(none)
`postgres`	Relational database (auth, KB metadata)	`5433`
`qdrant`	Vector database	`6333`
`redis`	Cache and task queue	`6379`
`frontend`	Pre-built React UI served by nginx	(no host port by default)

The first build may take **5–10 minutes** to download images and install dependencies.

Wait until you see log output indicating all services are ready. You can verify with:

```
# Check sync service health
curl http://localhost:8000/health

# Check query engine health
curl http://localhost:8001/health
```

Both should return `{"status": "ok"}` or similar.

Step 5: Start the Frontend

The recommended approach for local development is to run the Vite dev server outside Docker for fast hot-reloading.

Open a **new terminal window** and run:

```
cd frontend
npm ci
npm run dev -- --host
```

The Vite development server automatically proxies API requests:

- `/api/*` → `http://localhost:8000` (sync service)
- `/plugins/*` → `http://localhost:8000`
- `/sync` → `http://localhost:8000`

> **Alternative (Docker-only frontend):** If you prefer not to run Node.js locally, add the following to the `frontend` service in `docker-compose.yml` and then visit `http://localhost:5173`:
>
> ```
> ports:
>  - "5173:80"
> ```

Step 6: Access the Application

Open your browser and navigate to:

```
http://localhost:5173
```

You should see the InsightRAG landing page.

The following service dashboards are also available directly:

Service	URL
Sync Service API Docs	http://localhost:8000/docs
Query Engine API Docs	http://localhost:8001/docs
MCP Server	http://localhost:8002/mcp
Qdrant Dashboard	http://localhost:6333/dashboard

Step 7: First-Time Login

The database is automatically seeded with a default admin account. No manual user creation is needed.

1. Go to `http://localhost:5173`
2. Log in with the default admin credentials:

Field	Value
Email	`admin@example.com`
Password	`Admin123!`

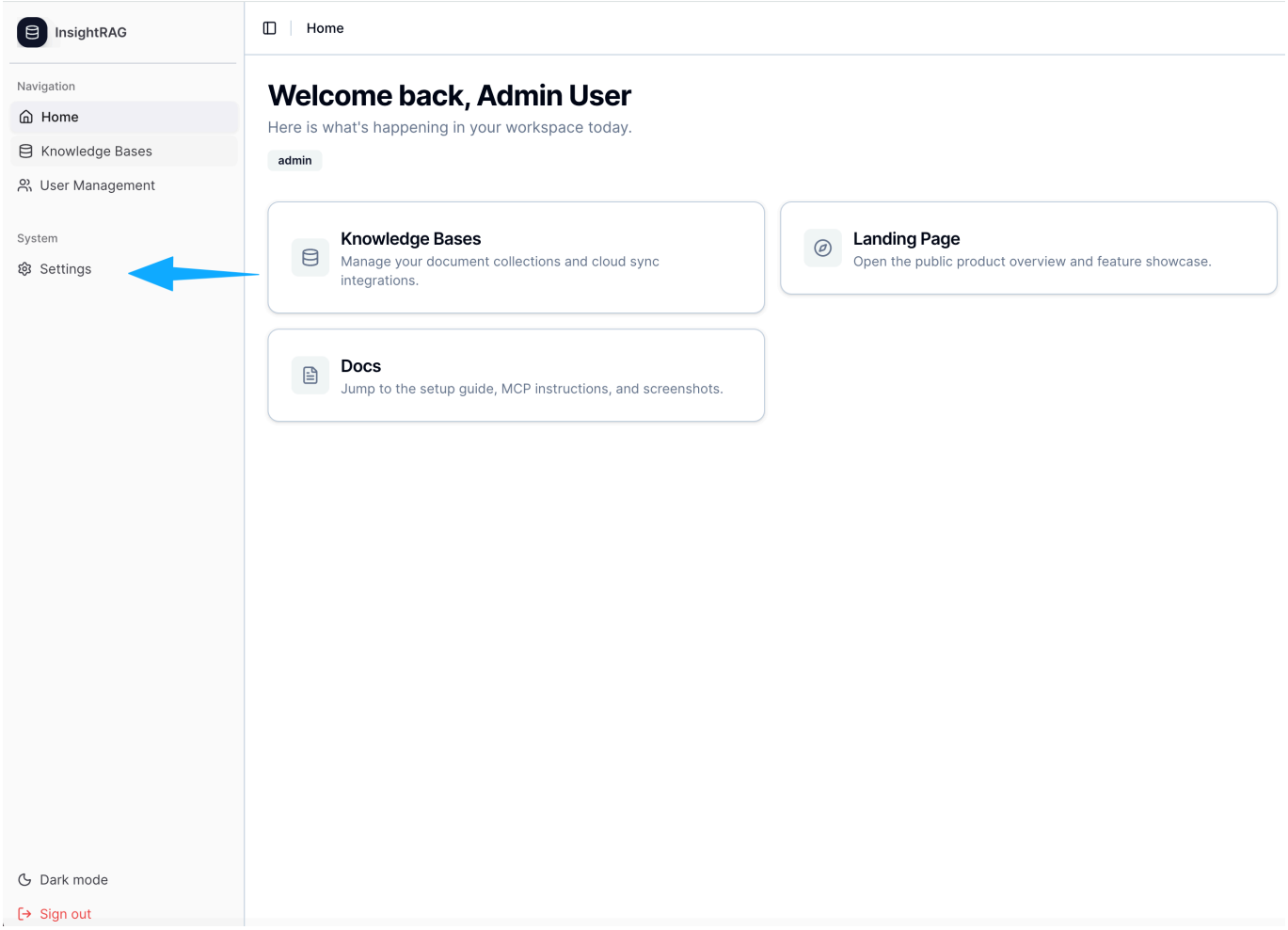
3. You will be redirected to the Knowledge Bases dashboard

> **Tip:** Once logged in, you can create additional users and assign roles (admin, developer, end_user) from the **User Management** page in the left sidebar.

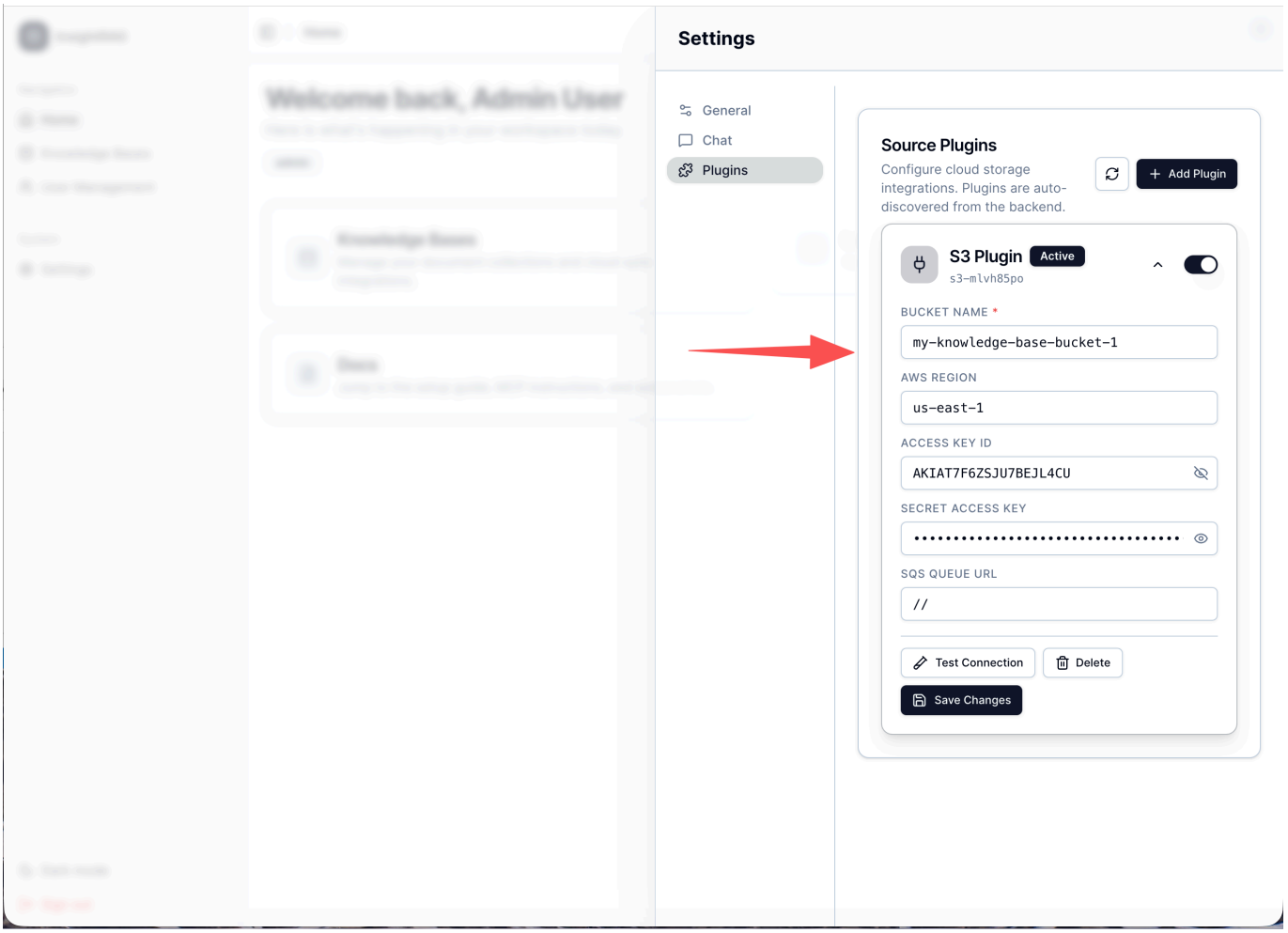
Step 8: Configure the S3 Plugin

Before syncing documents, you must configure a source plugin that tells the system where to fetch files from.

1. In the left sidebar, click **Settings**
2. Select the **Plugins** tab



3. Click **Add Plugin** (or the `+` button)
4. Fill in the plugin configuration:
   - **Name:** A descriptive name (e.g., `My S3 Source`)
   - **Module:** `app.plugins.s3`
   - **Class:** `S3Plugin`
   - **Active:** Toggle on
   - **Config → bucket_name:** Your S3 bucket name
   - **Config → region_name:** Your AWS region (e.g., `us-east-1`)
   - **Config → sqs_queue_url:** *(optional) but can not be empty* Your SQS queue URL for real-time sync



5. Click **Save**
6. Click **Test Connection** to verify the plugin can reach your S3 bucket

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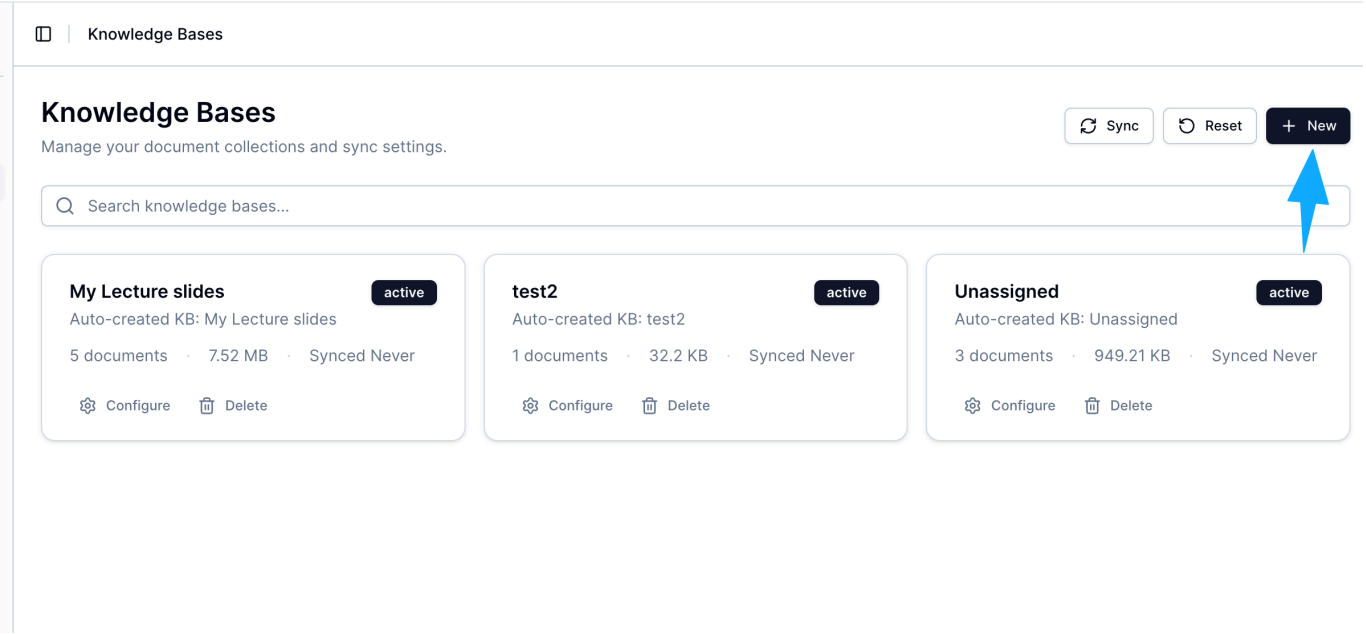
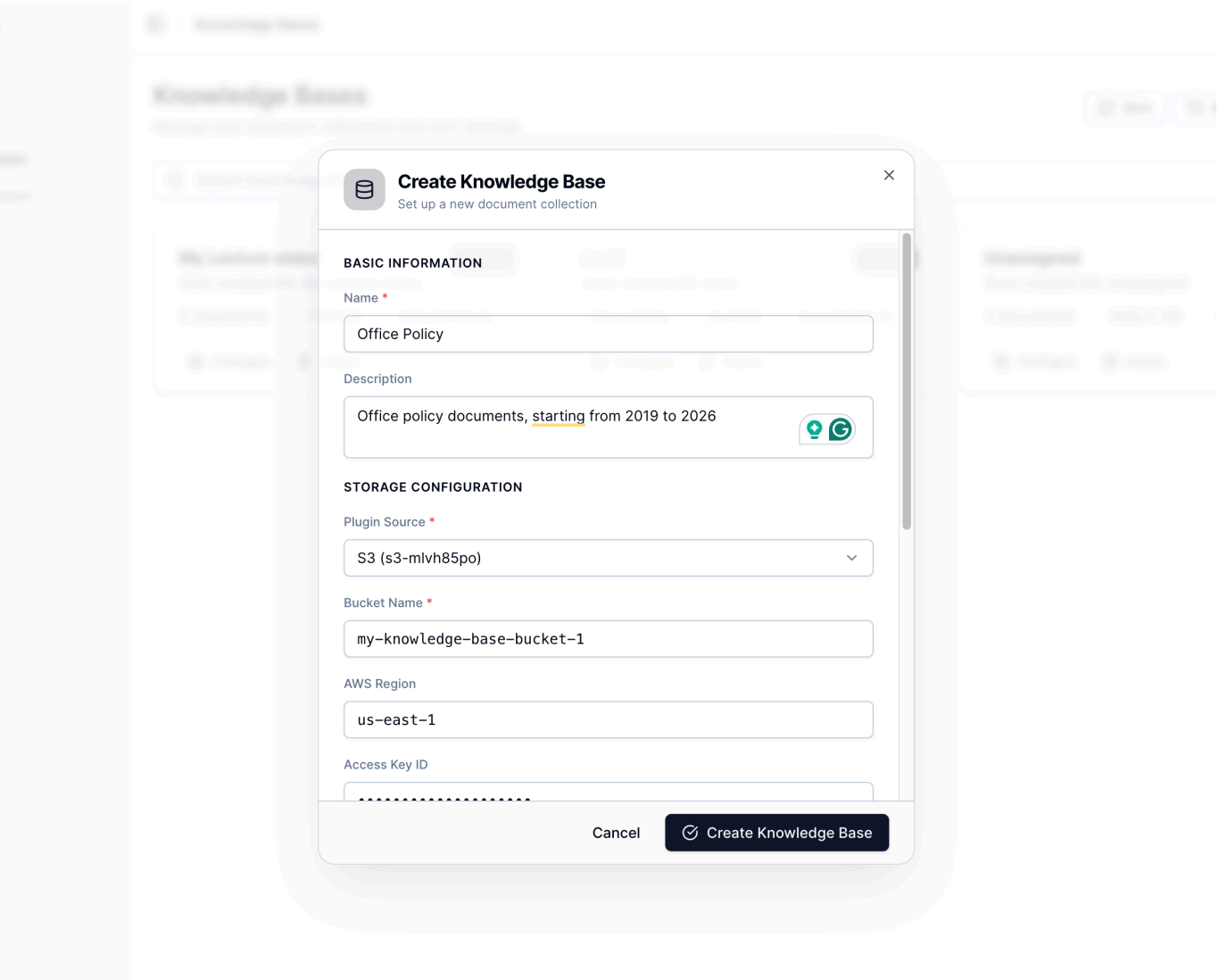
Step 9: Create a Knowledge Base

A Knowledge Base (KB) defines a logical collection of documents with its own processing configuration and routing rules.

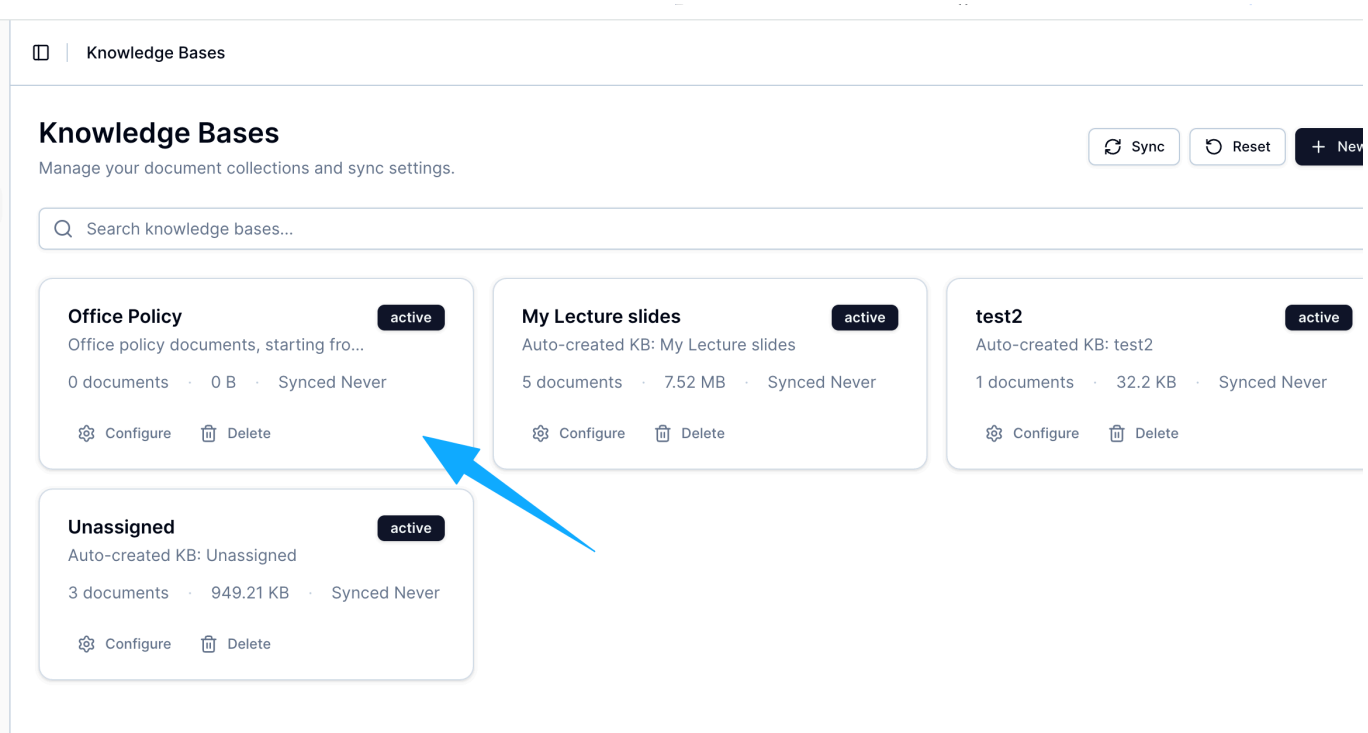
1. In the left sidebar, click **Knowledge Bases**
2. Click **New** (top right)



3. Fill in the form: - **Name:** A descriptive name (e.g., `Company Policies`) - **Description:** *(optional)* - **Storage Configuration → Plugin:** Select the S3 plugin you created - **Sync Folders:** *(optional)* Enter S3 path prefixes to route into this KB (e.g., `team-a/policies`). Leave empty to sync all files from the plugin. - **Chunk Size:** `512` (default — number of tokens per chunk) - **Chunk Overlap:** `50` (default — overlap between adjacent chunks)



4. Click **Create Knowledge Base**

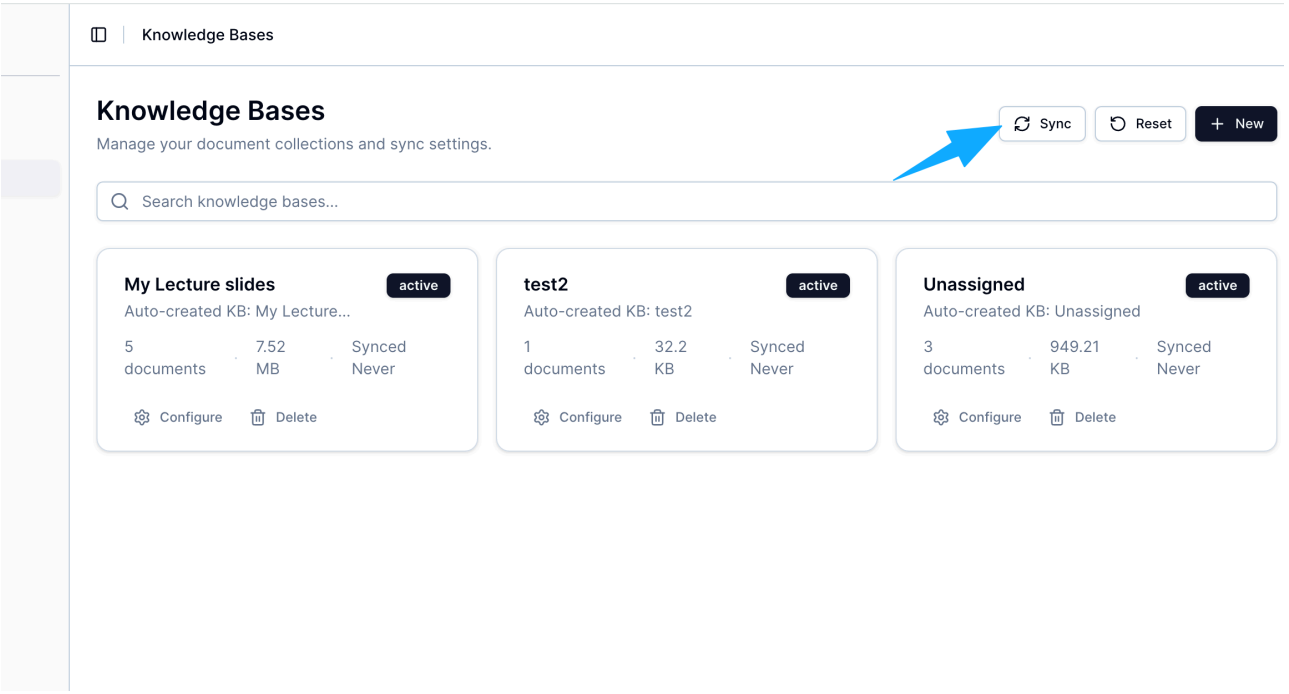


Routing logic: When the sync runs, files are routed to a KB based on path prefix matching. The longest matching prefix wins. A KB with no sync folders acts as a catch-all for its plugin.

Step 10: Sync Documents

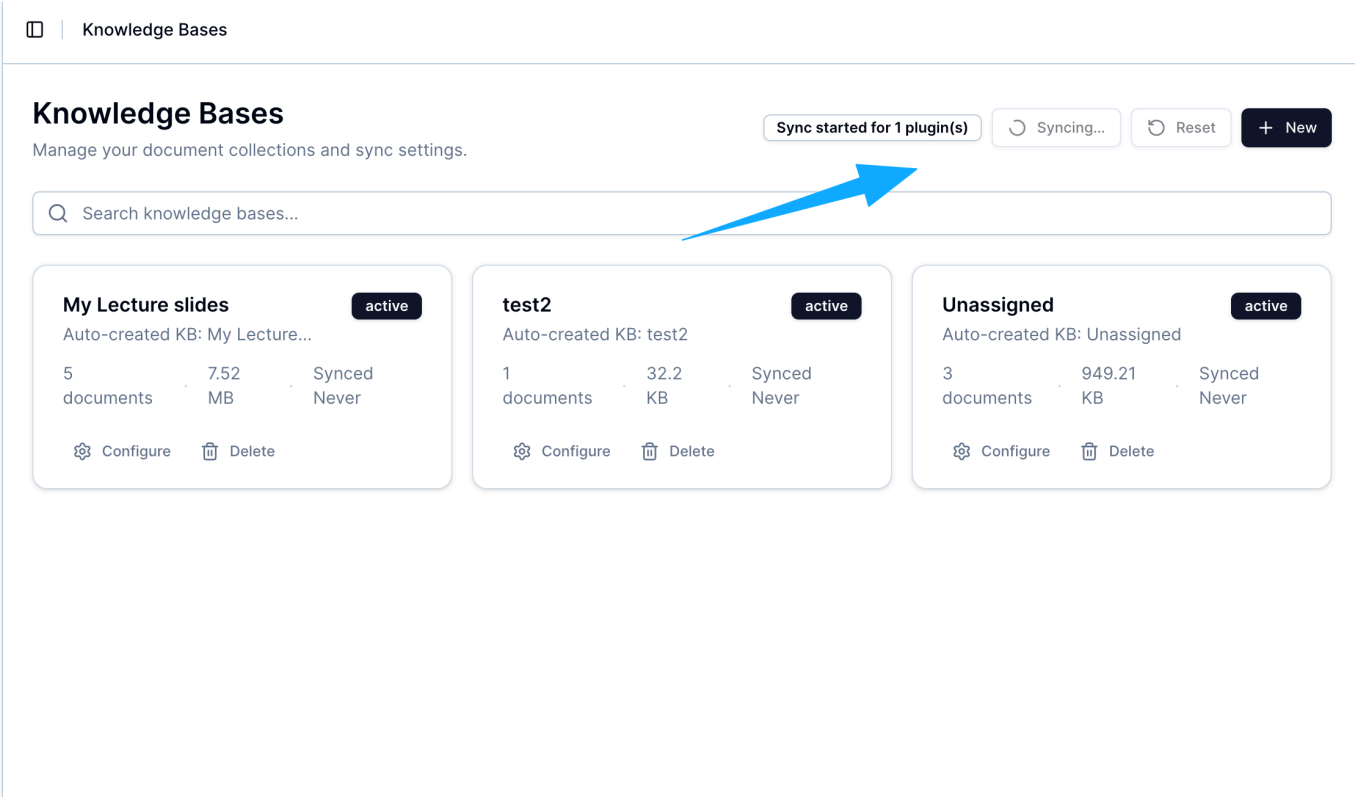
Once a plugin and KB are configured, trigger a sync to pull documents from S3.

1. On the **Knowledge Bases** page, click **Sync** in the top action bar



2. The button changes to **Syncing...** while the request is in flight

3. After the sync API call completes, a status badge appears



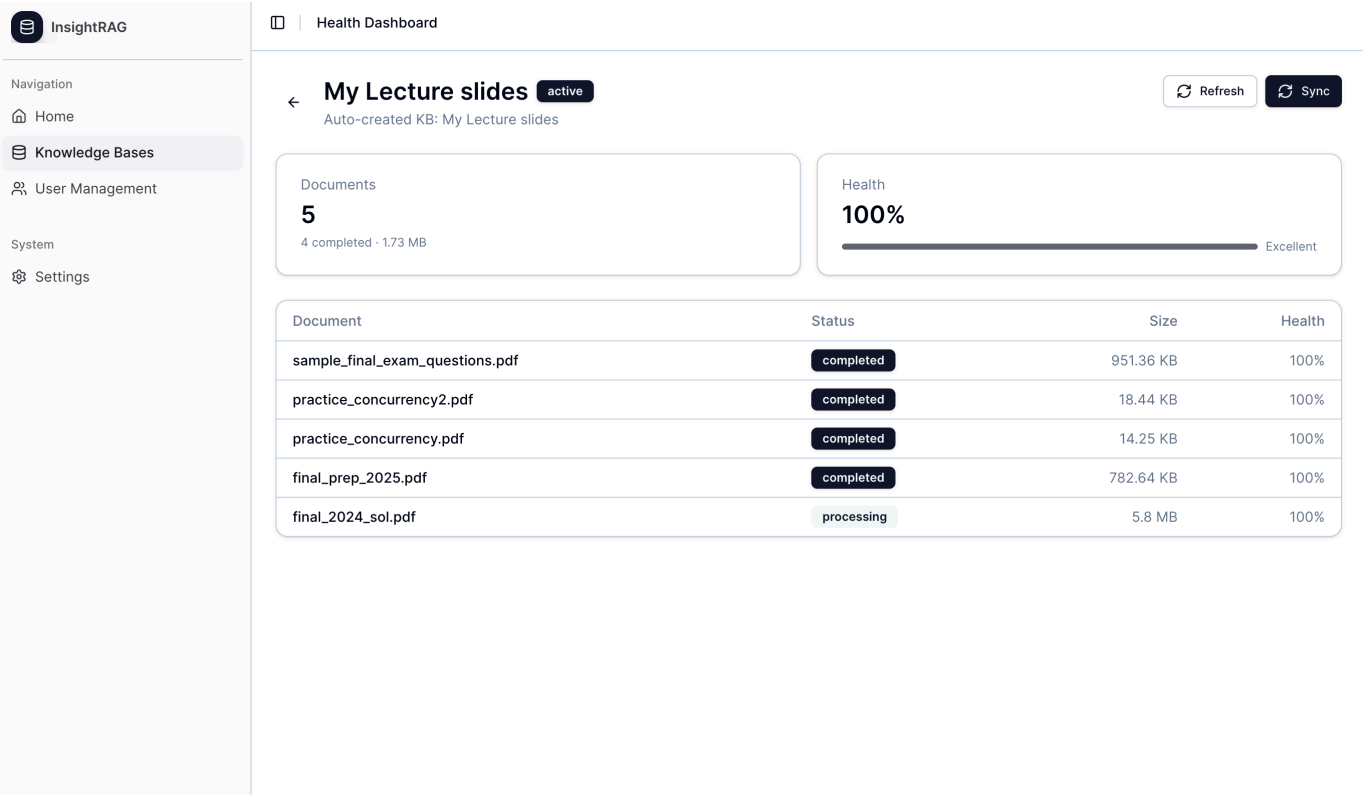
The sync process works as follows:

1. The sync service queries all active plugins for file listings
2. It compares against previously synced files (delta detection)
3. New or changed files are downloaded to a shared volume
4. Download tasks are queued in Redis for the document processing engine
5. The processing engine parses, chunks, embeds, and indexes each file into Qdrant

> **Background sync:** By default, a background sync runs every 300 seconds (configurable via `SYNC_INTERVAL_SECONDS` in `.env`). Set to `0` to disable automatic background sync.

> **Reset sync cache:** Click **Reset** to clear the sync delta cache. The next sync will re-process all files from scratch. Use with caution.

To check document processing status, click on a KB, then click on any document to view its status badge (`processing`, `completed`, or `failed`).



Step 11: Get Your Access Token (for MCP)

The MCP server requires a valid bearer token from the sync service auth system.

1. In the left sidebar, click **Settings**
2. Stay on the **General** tab
3. Find the **Session Tokens** section
4. Click **Copy** next to **Access Token**

Keep this token safe. It grants access to all knowledge bases associated with your account.

Step 12: Connect an MCP Client

InsightRAG exposes three MCP tools that AI assistants (Claude, etc.) can call:

Tool	Description
`search_knowledge_base`	Hybrid search (vector + keyword) across indexed documents
`get_available_collections`	List all knowledge bases accessible to the authenticated user
`list_kb_resources`	Full inventory of KBs and their documents with metadata

Option A: HTTP Transport (recommended)

The MCP server listens at `http://localhost:8002/mcp`.

Configure your MCP client with:

- **URL:** `http://localhost:8002/mcp`

- **Authentication:** Bearer token (paste the token from Step 11)

Claude Desktop example (`claude_desktop_config.json`):

```
{
  "mcpServers": {
    "insightrag": {
      "command": "curl",
      "args": [
        "-s", "-N",
        "-H", "Authorization: Bearer YOUR_ACCESS_TOKEN",
        "http://localhost:8002/mcp"
      ]
    }
  }
}
```

Option B: stdio Transport (local development)

```
./scripts/run_mcp_stdio.sh
```

The stdio wrapper keeps stdout clean for the MCP protocol.

Users > sherryxia > Library > Application Support > Claude > {} claude_desktop_config.json

```
1  {
2    "mcpServers": {
3      "knowledge-base": {
4        "command": "/Users/sherryxia/CPSC-319/OpenWebUI-Project/scripts/run_mcp_stdio.sh",
5        "args": [],
6        "env": {
7          "QDRANT_URL": "http://host.docker.internal:6333",
8          "DATABASE_URL": "postgresql://user:password@host.docker.internal:5433/openwebui"
9        }
10     }
11   },
12   "preferences": {
13     "localAgentModeTrustedFolders": [
14       "/Users/sherryxia/cpsc427",
15       "/Users/sherryxia/CPSC-319/OpenWebUI-Project"
16     ],
17     "coworkScheduledTasksEnabled": true,
18     "ccdScheduledTasksEnabled": true,
19     "sidebarMode": "epitaxy",
20     "coworkWebSearchEnabled": true,
21     "floatingAtollActive": true
22   }
23 }
```

Option C: OpenWebUI Integration

If you are using OpenWebUI:

1. Go to **Settings → Admin Settings → External Tools**
2. Click **+** → change type to **MCP** (not OpenAPI)
3. Set URL to `http://host.docker.internal:8002/mcp`
4. Set bearer auth with the token from Step 11
5. Enter an ID and name → click **Check Connection** → **Save**

> **Note:** If OpenWebUI only supports a single static bearer token, all OpenWebUI users will share that token's KB access scope.

---

5. How to Use the System

Once the system is running and you are logged in, the typical user workflow is:

Step-by-step workflow

```
Login → Configure Plugin → Create Knowledge Base → Sync Documents → Search
/ Query via MCP
```

1. Login

Go to `http://localhost:5173` (or the live URL) and sign in with the admin credentials from the Demo Access section above.

2. Configure a Source Plugin

Before syncing any documents, the system needs to know where to pull files from.

1. **Settings → Plugins → Add Plugin**
2. Choose **S3Plugin**, enter your AWS S3 bucket name and region
3. Click **Test Connection** to verify it works
4. Save

3. Create a Knowledge Base

A Knowledge Base (KB) is a named collection of documents with its own search index.

1. **Knowledge Bases → New**
2. Give it a name and select the plugin you configured
3. Optionally add **Sync Folders** (S3 path prefixes) to scope which files go into this KB
4. Click **Create Knowledge Base**

4. Sync Documents

Pull documents from S3 and process them into the search index.

1. On the **Knowledge Bases** page, click **Sync**

2. Wait for documents to process — click a document to check its status (`processing` → `completed`)
3. Once completed, the documents are chunked, embedded, and indexed in Qdrant

5. Browse & Monitor

- Click any Knowledge Base to see its documents
- Click a document to open the inline viewer, chunk browser, and retrieval analytics
- Visit the **Health Dashboard** (left sidebar) for KB-level metrics

6. Query via MCP (AI Assistant Integration)

1. **Settings → General → Session Tokens → Copy Access Token**
2. In your MCP client (e.g., Claude Desktop), configure:
   - **URL:** `http://localhost:8002/mcp`
   - **Authorization:** `Bearer <your_token>`
3. Use the `search_knowledge_base` tool to ask questions about your indexed documents

7. Manage Users (Admin only)

1. **User Management** (left sidebar) → list all users
2. Assign roles: `admin`, `developer`, or `end_user`
3. Each role has different levels of access to KBs and admin features

---

6. Feature Summary

6.1 Implemented Features

Frontend (React + Vite)

Feature	Description
Authentication	JWT login, token refresh, session expiry dialog, role-based UI
Landing page	Animated hero section, feature cards, demo gallery, stats section
Documentation pages	Quick start, plugin setup guide, MCP integration guide, team section
Knowledge Base dashboard	List all KBs, sync status badges, create/edit KBs
Document management	List documents per KB, delete, reassign, view inline
Inline document viewer	Renders PDFs, images, and text files directly in the browser
Document chunk browser	Browse individual chunks, view retrieval scores
Retrieval analytics	Per-document retrieval history with trend charts
Document heatmap	Per-page retrieval frequency heatmap for PDFs
Health dashboard	KB-level completion rates, chunk counts, retrieval statistics

Feature	Description
User management	Admin panel to list users and assign roles
Settings page	Plugin configuration, access token display, general preferences
Dark/light theme	Toggle with persistence in localStorage
Responsive design	Mobile-friendly layouts with Framer Motion animations

Supported Document Formats (27+)

Category	Formats
Documents	PDF (`.pdf`), Word (`.docx`, `.doc`), OpenDocument (`.odt`), Rich Text (`.rtf`), EPUB (`.epub`)
Spreadsheets	Excel (`.xlsx`, `.xls`), CSV (`.csv`), TSV (`.tsv`)
Presentations	PowerPoint (`.pptx`, `.ppt`)
Text & Markup	Plain text (`.txt`), Markdown (`.md`), HTML (`.html`), JSON (`.json`), XML (`.xml`), YAML (`.yaml`)
Images (OCR)	PNG (`.png`), JPEG (`.jpg`), Bitmap (`.bmp`), TIFF (`.tiff`), HEIC (`.heic`)

Processing Strategies

Strategy	Description
Semantic	Splits by sentence boundaries — best for narrative text
Auto	General-purpose PDF parsing, good default for most documents
Table Heavy	Preserves table structure in dense tabular PDFs
Multi-column	Handles newsletters, journals, and multi-column layouts
DataViz Heavy	Extracts charts and images from visual-heavy PDFs
Section Aware	Splits at headings, never mid-section
Table Preserving	Keeps each table intact — best for spreadsheets and CSVs
Slide Per Chunk	One chunk per slide — best for presentations

Backend Services

Feature	Description
Auth system	User registration, JWT login (HS256), refresh tokens, bcrypt password hashing
RBAC	Roles: admin, developer, end_user; permission tables

Feature	Description
Knowledge Base CRUD	Create, read, update, delete KBs with routing config
S3 plugin	List-objects sync with delta detection via `synced_file` table
Background scheduler	Configurable interval sync (default 300s)
Document processing pipeline	Parse → chunk → embed (sentence-transformers) → index (Qdrant)
Hybrid search	Vector + keyword retrieval with RRF (Reciprocal Rank Fusion) fusion
MCP server (HTTP)	Three tools, JWT-authenticated, per-user KB scoping
MCP server (stdio)	Local dev transport for stdio-based MCP clients
Retrieval tracking	All MCP searches logged to `document_retrieval_history`
API key management	Scoped API keys per user
Admin API	User listing, role assignment
CI/CD pipeline	GitHub Actions: lint, build, pytest, smoke test, deploy to DigitalOcean

---

6.2 Features Not Implemented

Feature	Notes
Chat interface	`Chat.tsx` component exists but is not wired into app routing and is not accessible from the UI
Additional sync plugins	Only AWS S3 is supported; no Google Cloud Storage, Azure Blob, local filesystem, or HTTP sources
Rate limiting	No rate limiting on any API endpoint
Observability / metrics	No Prometheus, Jaeger, or structured logging integration
Automatic re-processing after strategy override	Strategy override saves to the database but does not trigger the processing pipeline (marked as TODO in code)
Offline mode / graceful degradation	UI requires backend connectivity at all times

---

6.3 Partially Implemented Features

Feature	Status	Notes
Document strategy override	Partial	UI and API save the strategy to the database, but re-processing is not triggered. The `TODO` comment is at `backend/sync-service/routes/documents.py:435`.

Feature	Status	Notes
Chat page	Partial	`frontend/src/pages/Chat.tsx` contains OpenWebUI integration with model selection and markdown rendering, but no route exists in `App.tsx`.
Real-time sync via SQS	Partial	SQS integration is implemented in the S3 plugin and falls back gracefully to list-objects polling when SQS is not configured. Works end-to-end if SQS is set up.

---

7. Known Bugs and Limitations

Severity scale:

- **High:** Blocks core functionality or causes major failure
- **Medium:** Significantly affects usability or reliability, but the system still works
- **Low:** Minor issue, cosmetic issue, or edge case

Bug 1 — Document strategy override does not re-trigger processing `Medium`

What happens: When a user changes a document's chunking strategy in the UI (e.g., switching from "Auto" to "Table Heavy"), the new strategy is saved to the database but the document is not re-processed. The chunks in the search index still reflect the original strategy.

Impact on demo: The override control appears to work (no error shown), but search results will not reflect the new strategy until a future manual re-sync. This can confuse users who expect immediate re-indexing.

Workaround: Click **Reset** on the Knowledge Bases page and then **Sync** again to force full re-processing.

Bug 2 — Chat page is unreachable `Medium`

What happens: The `Chat.tsx` component exists in the codebase with OpenWebUI integration, model selection, and markdown rendering — but no route is defined for it in `App.tsx`. There is no way to navigate to the chat interface from the UI.

Impact on demo: The chat feature does not exist from the user's perspective. It is invisible in the running application.

Workaround: None. This feature was not completed in time for handover.

Bug 3 — Session expiry dialog broken state `Medium`

What happens: When a login session expires, the app shows a re-login dialog. If the user closes this dialog without logging in again, the app continues running but all API calls silently fail (401 errors). The UI shows no feedback.

Impact on demo: If a demo session expires mid-presentation, the app will appear broken. Refreshing the page and logging in again resolves it.

Workaround: Refresh the page if the app stops responding after an idle period.

Bug 4 — No rate limiting on sync/search endpoints `Medium`

What happens: Clicking **Sync** repeatedly in quick succession sends multiple sync requests to the backend. Each request creates a separate set of processing jobs in Redis, potentially causing duplicate document processing.

Impact on demo: Does not affect normal use. Only an issue if the sync button is clicked many times rapidly.

Workaround: Wait for the "Syncing..." state to clear before clicking Sync again.

Bug 5 — Sync Reset is a global operation `Low`

What happens: Clicking **Reset** clears the sync delta cache for all knowledge bases and all plugins at once. If a sync is running concurrently, it may lead to files being re-queued unexpectedly.

Impact on demo: Low risk in single-user demo scenarios. Only relevant in a multi-user concurrent environment.

Bug 6 — Generic S3 error messages `Low`

What happens: When a document's S3 presigned URL cannot be generated (e.g., due to expired credentials or missing permissions), the UI shows "Failed to generate S3 URL" with no detail about the root cause.

Impact on demo: The inline document viewer may fail to load PDFs or images. The document list still shows correctly.

Workaround: Verify AWS credentials in `.env` and confirm the IAM user has `s3:GetObject` permission.

Bug 7 — No frontend tests `Low`

What happens: There are no unit or integration tests for any React component. Correctness relies entirely on manual testing.

Impact on demo: Does not affect the running application. A risk for future development — regressions may not be caught.

Bug 8 — Minimal backend test coverage `Low`

What happens: Only the sync-service has tests (3 test cases: plugin manager, password hashing, JWT). The query engine, MCP server, and document processing engine have no tests.

Impact on demo: Does not affect the running application. A risk for future development.

Bug 9 — KB health score misleading during processing `Low`

What happens: The health score for a Knowledge Base is calculated only from documents with `completed` status. While documents are still processing, the score may appear lower than the final expected value.

Impact on demo: After a sync, wait a few minutes for processing to finish before presenting health metrics.

Bug 10 — No file size limits on ingestion `Low`

What happens: There is no file size check before a document is queued for processing. Very large files (e.g., a 500 MB PDF) may cause the document-processing-engine to time out silently.

Impact on demo: Not an issue with normal-sized documents (under ~50 MB). Avoid uploading unusually large files during demos.

---

8. Codebase Structure

```
OpenWebUI-Project/
├── frontend/                          # React + Vite + TypeScript UI
│   ├── src/
│   │   ├── pages/                     # Page-level components
│   │   │   ├── KnowledgeBases.tsx     # KB list + sync controls
│   │   │   ├── DocumentDetails.tsx    # Document viewer + analytics
│   │   │   ├── HealthDashboard.tsx    # KB health metrics
│   │   │   ├── Settings.tsx           # Plugin config + tokens
│   │   │   ├── UserManagement.tsx     # Admin user panel
│   │   │   ├── Docs.tsx               # Documentation pages
│   │   │   ├── Landing.tsx            # Landing page
│   │   │   └── Chat.tsx               # (partial) Chat interface
│   │   ├── components/                # Reusable UI components
│   │   ├── App.tsx                    # Root component + routing
│   │   └── main.tsx                   # Entry point
│   ├── package.json                   # Dependencies + scripts
│   └── vite.config.ts                 # Vite config + API proxy
│
├── backend/
│   ├── sync-service/                  # FastAPI: auth, KB CRUD, plugins,
sync
│   │   ├── app/
│   │   │   ├── plugins/               # Plugin implementations
│   │   │   │   ├── interface.py       # SourcePlugin base class
```

```
│   │   │   │   └── s3.py              # AWS S3 plugin
│   │   │   └── sync_service.py        # Sync orchestration logic
│   │   ├── routes/                    # API route handlers
│   │   │   ├── auth.py                # Login, logout, refresh
│   │   │   ├── knowledge_bases.py     # KB CRUD
│   │   │   ├── documents.py           # Document management
│   │   │   ├── plugins.py             # Plugin management
│   │   │   └── users.py               # User/admin management
│   │   ├── main.py                    # FastAPI app + sync scheduler
│   │   └── AWS_SETUP.md               # AWS S3/SQS setup guide
│   │
│   ├── document-processing-engine/    # Celery worker:
parse/chunk/embed/index
│   ├── query-engine/                  # FastAPI: search API for the UI
│   ├── mcp-server/                    # FastMCP: MCP tools + JWT auth
│   ├── database/                      # PostgreSQL schema + seeds
│   │   ├── init.sql                   # Full schema (auto-runs on postgres
start)
│   │   ├── seed_documents.sql         # Demo document data
│   │   └── reset_all.sql              # Full database reset
│   └── shared/                        # Shared Python utilities
│
│
├── docs/                              # Additional documentation
│   ├── plugin-setup.md                # Plugin configuration guide
│   ├── plugin-development.md          # Writing new sync plugins
│   ├── setup-openwebui.md             # OpenWebUI + MCP integration
│   ├── migration.md                   # Database migration notes
│   └── unit-tests.md                  # Running tests locally
│
├── scripts/
│   └── run_mcp_stdio.sh               # stdio MCP transport wrapper
│
├── diagrams/                          # Architecture diagrams
├── docker-compose.yml                 # Local dev stack (all 8 services)
├── .env.example                       # Environment variable template
├── .github/workflows/ci-cd.yml        # CI/CD: lint → test → build →
deploy
├── README.md                          # Quick start reference
├── skill.md                           # MCP tool reference for LLM agents
└── HANDOVER.md                        # This document
```

---

9. Future Work & How to Continue

This section is intended for any developer or team who picks up this project after handover.

Where to start

1. **Read the codebase overview** in Section 8 above — the `backend/sync-service/` and `frontend/src/` directories are the core of the system

2. **Run locally** following the setup guide (Section 4) — the full stack starts with `docker compose up --build`
3. **Read `docs/plugin-development.md`** before adding new sync sources
4. **Check the `TODO` in `backend/sync-service/routes/documents.py:435`** — this is the highest-priority incomplete item

Suggested next steps

High priority

Task	Description
Fix strategy override	After a user changes a document's chunking strategy, trigger a re-processing job (the DB save already works; only the Celery task dispatch is missing)
Wire up Chat page	Add a route for `Chat.tsx` in `App.tsx` and test the OpenWebUI MCP integration end-to-end
Add rate limiting	Add `slowapi` or similar middleware to the sync-service FastAPI app to throttle `/sync` and search endpoints

Medium priority

Task	Description
Add more sync plugins	Implement `GCSPlugin` (Google Cloud Storage) or `AzureBlobPlugin` by following the `SourcePlugin` interface in `backend/sync-service/app/plugins/interface.py`
Add observability	Integrate Prometheus metrics + Grafana dashboard for request rates, processing queue depth, and search latency
Expand test coverage	Add pytest tests for the query engine and MCP server; add Playwright tests for critical frontend flows
File size validation	Add a file size check before queuing documents for processing to prevent worker timeouts

Low priority / nice to have

Task	Description
Streaming search results	Return MCP search results as a stream rather than a single response
Multi-tenant KB sharing	Allow a KB to be shared across multiple users with different permission levels
Structured logging	Replace `print()`/basic logging with structured JSON logs for easier log aggregation
Session handling hardening	Fix the session-expiry broken-state bug (Bug 3 above)

How to add a new sync plugin

1. Create a new file in `backend/sync-service/app/plugins/` (e.g., `gcs.py`)
2. Implement a class that inherits `SourcePlugin` from `interface.py`
3. Implement the `sync()` method to yield `FileEvent` objects
4. Restart the sync-service — the plugin is auto-discovered
5. Configure an instance via **Settings → Plugins** in the UI

See `docs/plugin-development.md` for a full walkthrough.

Key contacts

Person	Role
Jaskirat Gill	Backend architecture, infrastructure, CI/CD
Sherry Rui Xia	Frontend, documentation pages

---

10. Troubleshooting

Docker containers fail to start

Symptom: One or more services exit immediately after `docker compose up`.

Fix: Check logs with `docker compose logs <service-name>`. Common causes:

- Port conflicts: ensure ports 8000, 8001, 8002, 5433, 6333, 6379 are free on your host
- Missing `.env` variables: ensure `.env` exists with all required values

No plugins found in Settings → Plugins

Symptom: The plugin dropdown in Settings is empty.

Fix: Confirm that `backend/sync-service/app/plugins/s3.py` exists and that the `S3Plugin` class inherits from `SourcePlugin` (`backend/sync-service/app/plugins/interface.py`). After fixing, restart the sync-service container.

Plugin test connection fails

Symptom: Clicking **Test Connection** returns an error.

Fix:

- Confirm your AWS credentials in `.env` are correct
- Verify the S3 bucket exists and your IAM user has `s3:ListBucket` permission
- Use `POST /plugins/{plugin_id}/test` in the API docs (`http://localhost:8000/docs`) for detailed error output

Sync runs but no documents appear in the KB

Symptom: Sync completes but the Knowledge Base shows 0 documents.

Fix:

1. Check that the KB's **Storage Configuration → Plugin** matches the plugin instance ID
2. Verify that the **Sync Folders** path prefixes match the actual S3 key prefixes (e.g., `team-a/` must match the S3 object keys)
3. Leave **Sync Folders** empty to route all files from the plugin into this KB

Documents stuck in "processing" status

Symptom: Documents show `processing` for more than a few minutes and never complete.

Fix:

- Check the document-processing-engine logs: `docker compose logs document-processing-engine`
- Ensure Redis is healthy: `docker compose logs redis`
- Very large files may time out during parsing; check for timeout errors in the logs

MCP authentication fails

Symptom: MCP client returns `401 Unauthorized` or `403 Forbidden`.

Fix:

- Ensure you copied the **Access Token** from Settings → General, not an API key
- The token must be passed as `Authorization: Bearer <token>` in the HTTP header
- Tokens expire; log in again to get a fresh token if needed

Frontend proxy errors (404 on /api routes)

Symptom: API calls from the UI return 404 when running the Vite dev server.

Fix: Confirm the backend Docker containers are running on the expected ports. Check `vite.config.ts` for the proxy target — it should point to `http://localhost:8000`.

InsightRAG — CPSC 319 Final Project Handover | April 2025
GitHub: https://github.com/jaskirat-gill/InsightRAG